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ABSTRACT

B-field measurements are crucial for the study of high-temperature and high-energy-density plasmas. A successful diagnostic method, ion deflectometry (radiography), is commonly employed to measure MGauss magnetic fields in laser-produced plasmas. It is based on the detection of multi-MeV ions, which are deflected in B-fields and measure their path integral. Until now, protons accelerated via laser–target interactions from a point-like source have been utilized for the study of Z-pinch plasmas. In this paper, we present the results of the first Z-pinch-driven ion deflectometry experiments using MeV deuterium beams accelerated within a hybrid gas-puff Z-pinch plasma on the GIT-12 pulse power generator. In our experimental setup, an inserted fiducial deflectometry grid (D-grid) separates the imploding plasma into two regions of the deuteron source and the studied azimuthal B-fields. The D-grid is backlighted by accelerated ions, and its shadow imprinted into the deuteron beams demonstrates ion deflections. In contrast to the employment of the conventional point-like ion source, in our configuration, the ions are emitted from the extensive and divergent source inside the Z-pinch. Instead of having the point ion source, deflected ions are selected via a point projection by a pinhole camera before their detection. Radial distribution of path-integrated B-fields near the axis (within a 15 mm radius) is obtained by analysis of experimental images (deflectograms). Moreover, we present a 2D topological map of local azimuthal B-fields B(r,z) via numerical retrieval of the experimental deflectogram.

I. INTRODUCTION

Magnetic fields have a crucial role in the behavior of high-temperature and high-energy-density plasmas. In Z-pinch discharges, the B-field and current topology are essential for studying the dynamics of imploding plasmas and energy coupling to the plasma. Complex dynamics of the imploding plasma are investigated using numerical simulations, which must be verified against some experimental data. Conventionally, there are several methods for the measurements of the magnetic fields in Z-pinch plasmas. However, measurements of their spatial distribution are a difficult task for any of them. We are referring to these B-field measurement methods: (1) Faraday rotation; (2) B-dot probes; (3) Zeeman broadening, and all have some limitations.

In the first method, the Faraday rotation, a plane of polarization of an electromagnetic (EM) wave, rotates while propagating through the plasma parallelly to B-fields $B_{\|}$. An angle of the rotation β is proportional to $\int n_e(s)B_{\|}(s)\mathrm{d}s$, where $n_e(s)$ and $B_{\|}(s)$ are the electron density and the parallel B-fields at each point s along the path. Due to the necessity of non-negligible electron density, this method cannot be used in near-vacuum environments. A proper determination of an averaged B-field requires specific knowledge of the electron density profile $n_e(s)$. It can be obtained either by simultaneous interferometric measurements, which bring an additional source of error or by the employment of a magneto-optical fiber with known length and density,[1,2] which must not be invasive to the measured plasmas. On the other hand, diagnostic beams can be used only in environments with densities lower than the critical density, which obstructs measurements in dense plasmas close to the Z-pinch axis during a stagnation. However, using ultraviolet laser diagnostics at the wavelength of 266 nm,[1] it has been possible to study plasmas up to densities of $\sim 10^{20}\,\mathrm{cm}^{-3}$.

B-dot probes measure the temporal derivative of the B-fields by a coil inserted into the plasma. For high-temperature and high-density plasmas of the Z-pinch, these probes may be intrusive. Thus, they are usually used to study B-fields at peripheral regions of the Z-pinch or placed in front of the imploding plasma. Measurements of B-fields in hot dense plasmas close to the Z-pinch axis are indirect. They require estimating the B-field evolution or extrapolation via numerical simulations.[3,4] To minimize the influence of the inserted probe, a micro-B-dot probe has been developed at Cornell.[5,6] A set of B-dot probes is needed to measure the B-field distribution, which makes this method more perturbative.

In the presence of magnetic fields, the emission lines split due to the Zeeman effect. The shift of lines in the spectral emission profile is proportional to the magnetic strength. High temperature and high density of the plasma cause Stark and Doppler broadening, which smear out the fine structure of the Zeeman splitting. Therefore, the employment of different components of the same multiplet is required, then the exact spectral measurements rely on the line positions rather than their widths. Only certain lines of sight are possible for the measurements, and thus, the study of the spatial distribution of the magnetic field is difficult. However, the B-field profile has been measured for radius $r > 7\,\mathrm{mm}$[7] and, recently, using two-polarization method.[8,9] The latter experiments have been performed in Z-pinch oxygen plasmas. Employment of the Zeeman method in low-Z plasma (e.g., hydrogen or deuterium) may be problematic.

Another alternative method of B-field measurements in hot dense plasmas has risen in the last two decades. It is an *ion deflectometry* (or more general name, an *ion radiography*). This diagnostic method is based on detecting ion beams deflected by either perpendicular electric or magnetic fields. As ions propagate through these fields, their trajectories bend due to the Lorentz force. When recorded by an ion detector, these deflected beams create a distorted image, a *deflectogram*. Ion displacements in the deflectogram are proportional to the path integral of these fields, that is, $\int \boldsymbol{E}_{\perp} dL$ for electric fields and $\int \boldsymbol{B} \times d\boldsymbol{L}$ for magnetic fields, which are main quantities inherently coupled to this diagnostics. This paper considers only ion deflections in the azimuthal magnetic fields B_{φ} of the Z-pinch. The path integral of B-fields $\int \boldsymbol{B} \times d\boldsymbol{L}$ will be discussed in Sec. II. The ion radiography has been developed in the laser–plasma community as methods for measurements in hot dense plasmas using multi-MeV protons.[10,11] In the past years, the laser-accelerated proton beams allowed radiographic measurements of E- or B-fields in experiments related to plasma jets,[12–15] laser-target interactions accelerating multi-MeV protons,[16–18] laser-driven capacitor coils,[19–22] Weibel instability,[23,24] plasma flows in hohlraums,[25] imploding inertial confinement fusion (ICF) capsules,[26–28] propagation of laser-driven EM pulses in metallic wires,[29,30] magnetic reconnections,[31–35] and self-generated B-fields in laser-produced plasma bubbles[36–40] with excellent μm-spatial and *ps*-temporal resolution. Moreover, ion deflections can be also exploited for field measurements in other types of plasmas. In tokamak plasmas, the heavy ion beam probe (HIBP) diagnostic, commonly used for electron density and plasma potential measurements, utilizes deflections of heavy ions for the investigation of poloidal B-field fluctuations.[41–46]

In Z-pinch plasmas, there were only a few experimental measurements of magnetic fields using proton beams. In 2012, dynamic electric fields and return currents were measured in laser-driven Z-pinch plasmas created by irradiation of a boron stalk at the Omega facility.[47] In 2014, the proton deflectometry was tested in azimuthal vacuum magnetic fields driven by Zebra, a MA pulsed-power Z-pinch device in Nevada Terawatt Facility (NTF).[48] Moreover, experiments with radial foil loads were performed to demonstrate the feasibility of the proton deflectometry in the pulse power plasmas. There were experiments with other loads in NTF,[49] including a hybrid X-pinch and cylindrical wire array configuration. The proton deflectometry has also been considered for Z-machine.[50] However, numerical simulations showed that 4.5-GeV probe protons would be necessary to radially traverse the azimuthal magnetic fields near the axis of the imploding 20-MA liner, which is unrealistic. Instead, the deflectometry was suggested for fringe B-field measurements, but it would still necessitate at least 30 MeV protons. These simulations pointed out two main issues of employment of the ion deflectometry in azimuthal Z-pinch-like magnetic fields.

The first one is a source of ion beams with sufficient momentum, so the Larmor radius of ions is large enough compared to the size of the plasma, and hence, the ion deflections are manageable. It is convenient for experiments in dense plasmas to utilize light ions, for example, protons or deuterons, but then their necessary energy must be in terms of MeV. In all experiments mentioned in the last paragraph, multi-MeV protons were produced separately via laser–target interactions by either fusion reactions (DD or D^2He) or the target normal sheath acceleration (TNSA) mechanism. Therefore, these experiments required a high-intensity short-pulse laser system connected to the Z-pinch assembly. However, such laser systems are only available at a few Z-pinch facilities. The second point in question is the direction of the probing ion beams. In all previous experiments, the proton beams were emitted into azimuthal B-fields radially.[47,48] In this configuration, ions must traverse the peripheral B-fields, which dominantly affect ion trajectories, so the spatial B-field distribution measurements near the axis are nearly impossible. An axial deflectometry allows mapping of the B-fields near the axis. However, if ions travel upstream, in the opposite direction as the current, their initial divergence is extended by the defocusing deflections in azimuthal B-fields. In strong B-fields, the deflectograms may become large to be recorded in the plane detector located far from the plasma.[51] If the ion focusing direction is chosen, the deflected ion trajectories may cross each other, which may cause problems for the deflectogram interpretation.

This paper introduces another approach to the deflectometric measurements of azimuthal B-field of the Z-pinch in an unconventional experimental setup. We studied an acceleration mechanism of multi-MeV hydrogen ions in Z-pinch plasma for a few past years in experiments on the 4.7-MA GIT-12 pulse generator in Tomsk, Russia. The maximum energy of these hydrogen ions (mostly deuterons) reaches up to 55 MeV.[52] These multi-MeV deuteron beams are accelerated downstream inside the Z-pinch so that we can investigate the distribution of the azimuthal B-fields of the Z-pinch close to the axis ($<15\,\mathrm{mm}$) without a high-intensity laser system. Although the ion source region is extensive (i.e., not point-like) and its properties are still subject to our study, B-field measurements are possible because the point-like projection selects the deflected deuterons via a pinhole camera. We can avoid undesirable crossing of the ion trajectories in the azimuthal B-fields because the small pinhole on top of the ion detector acts as a well-defined crossing point of ion trajectories. Because the probing deuterons originate from within the Z-pinch itself, we call our method a *Z-pinch-driven ion deflectometry*.

This paper is structured into sections and subsections as follows: In Sec. II, we discuss the path integral of B-fields $\int \boldsymbol{B} \times d\boldsymbol{L}$ and its radial component $\int B_{\varphi} dz$ along ion trajectories, which are crucial quantities for the ion deflectometry B-field measurements. In Sec. III, we describe experiments on GIT-12 and the analysis techniques for estimating averaged B-fields $\overline{B_{\varphi}}$ from the measured ion displacements, and for the examination of the deflectogram structure. This section introduces the three vital elements of our unique diagnostics that enable the deflectometric measurements of azimuthal B-fields in Z-pinch plasmas. In Subsection III A, we describe the first two, which are coupled to the Z-pinch. The first one is an alternative ion source for the backlighting by employment of the deuteron beams accelerated inside the Z-pinch. The second one is a fiducial deflectometry grid (*D-grid*), which we insert into the anode–cathode gap. It separates the ion source from the rest of the imploding plasma and its distorted shadow captures ion deflections in the azimuthal B-field below the D-grid. Conventional ion sources for the deflectometry are point-like, but in our experiments, deuterons are accelerated in a broad region above the D-grid. However, the deflectometric measurements are still possible if we detect the deflected ions via a point-like projection (e.g., through a small pinhole). In Subsection III B, we explain the deflectometry setup and introduce the last key component of our diagnostics, which is a distant pinhole camera for the detection of deflected deuteron beams. In Subsection III C, we propose an analytic method for the measurement of the averaged B-fields $\overline{B_\varphi}$, coupled to the path-integrated B-field $\int B_\varphi dz$. In Subsection III D, we use our ion-tracking simulations and illustrate four types of characteristic distortions of synthetic deflectograms, which are coupled to four current density distributions.

In Sec. IV, we discuss our experimental results and analysis. In Subsection IV A, we present the first-ever experimental Z-pinch-driven deflectograms. The quality of these data varies, and thus, the deflectograms provide different amounts of information about the B-fields. In Subsection IV B, we show that even low-quality deflectograms without a distorted D-grid shadow allow us to estimate the total current in the Z-pinch during the ion imaging. In Subsection III D, we investigate the formation and the structure of the D-grid patterns in the synthetic deflectograms via our ion-tracking simulations. In Subsection IV C, we analyze three selected deflectograms with recognizable D-grid shadows and reconstruct 2D (x-y) maps of the averaged B-fields, assuming the axially uniform B-fields. In Subsection IV D, we carry our analysis of one experimental deflectogram even further. Using an experimental deflectogram and a side view, namely a soft x-ray (SXR) image, of the imploding plasma captured during the ion imaging, we numerically reconstruct a 2D (r,z) topological map of the local B-fields and the current. At the end of this paper (Sec. V), we conclude our experimental results and outline our plans to improve our future experiments employing ion deflectometry for B-field measurements in Z-pinch plasmas.

II. DEFLECTION EQUATION AND PATH-INTEGRATED MAGNETIC FIELD

Ion deflectometry is used for measurements of E- and B-fields. In this paper, we exclude electric fields, and we assume a highly conducting Z-pinch plasma with solely azimuthal magnetic fields $\boldsymbol{B}_\varphi$, which are nearly stationary for the passing ions. Let the energy of ion beams be sufficiently high, so the collisional scattering and the stopping power are low even in high-density plasmas. Furthermore, let the ion beam has a space-charge potential negligible compared to the beam energy E so that the ions act as individual test particles.

A cartoon of the conventional experimental setup of the ion deflectometry is illustrated in Fig. 1. Ions are emitted axially toward the plasma from a divergent point-like source, so the ion beams are

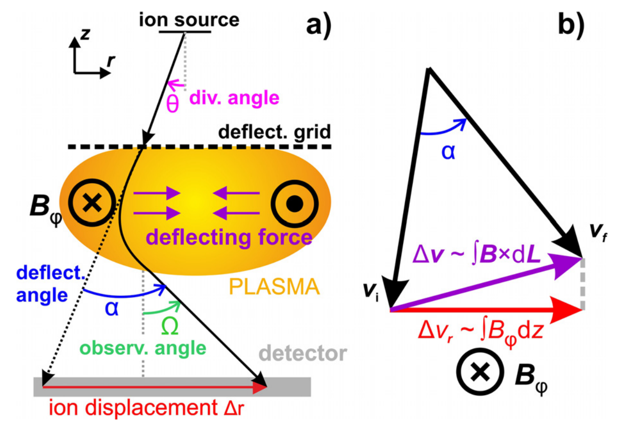

FIG. 1. (a) Scheme of a typical experimental setup of the ion deflectometry. Ions are emitted into the plasma through the deflectometry grid (D-grid), which splits the beams into the beamlets. An image of the ions deflected in the azimuthal B-field in the plasma is recorded by an ion detector. By measuring deflection angles α from ion displacements in the detector Δr, we estimate the path-integrated B-field. **(b)** Deflections of the ion in the perpendicular B-fields cause a rotation of the ion velocity vector , which can be characterized by a change of the its direction $\Delta \boldsymbol{v}$ from its initial velocity $\boldsymbol{v}_i$ to its final velocity $\boldsymbol{v}_f$ by the deflection angle α. The vector $\Delta \boldsymbol{v}$ is proportional to the path-integrated B-fields $\int \boldsymbol{B} \times d\boldsymbol{L}$ and its radial component Δv_r to $\int B_\varphi$dz. Because the ion energy is conserved during the motion in B-fields, the magnitude of the velocity remains the same, that is, $v = ||\boldsymbol{v}_f|| = ||\boldsymbol{v}_i||$.

laminar. Before entering the plasma, the ions pass through a fiducial deflectometry grid (*D-grid*). A shadow of the grid is imprinted into the ion beams, which split into beamlets. In the plasma, the ions are deflected by a deflection angle α due to magnetic fields. Finally, the deflected ions reach an ion detector, usually a filtered stack of radiochromic films (RCF) or CR-39 track detectors, and create a distorted image of the D-grid shadow, a *deflectogram*.

When the ions are going through the magnetic fields, the Lorentz force changes directions of their velocity vectors. The initial velocity $\boldsymbol{v}_i$ is rotated to its final velocity $\boldsymbol{v}_f$. We can characterize the rotation as a velocity shift by a vector $\Delta \boldsymbol{v} = \boldsymbol{v}_f - \boldsymbol{v}_i$, for which we can write

$$\Delta \boldsymbol{v} = -\frac{Q}{m}\int \boldsymbol{B} \times \boldsymbol{v} \mathrm{d}t = -\frac{Q}{m}\int \boldsymbol{B} \times d\boldsymbol{L}, \tag{1}$$

where Q and m are the ion charge and mass, and $\boldsymbol{v}\mathrm{d}t = d\boldsymbol{L}$ is an element of the ion path. During the magnetic deflections, the magnitude of the ion velocity vector remains the same $v = ||\boldsymbol{v}_f|| = ||\boldsymbol{v}_i||$. The vector integral $\int \boldsymbol{B} \times d\boldsymbol{L}$ denotes the path-integrated B-field, which is a crucial quantity describing the cumulative effect of magnetic fields on deflecting ions along the ion path. It is inherently coupled with a *deflection angle* α by which the ions are deflected, and their velocities are rotated during their deflections. The deflection angle α is related to a strength of the path-integrated B-field $||\int \boldsymbol{B} \times d\boldsymbol{L}||$ by a *deflection equation*[51]

$$\left|\sin\left(\frac{\alpha}{2}\right)\right| = \frac{||\Delta v||}{2v} = \frac{Q}{2mv}\left\|\int \boldsymbol{B} \times d\boldsymbol{L}\right\|. \tag{2}$$

The path-integrated B-field $\int \boldsymbol{B} \times d\boldsymbol{L}$ and the deflection angle α have similar importance for the ion deflectometry as do the integral $\int n_e(s) B_{\|}(s) \mathrm{d}s$ and the rotation angle β for the Faraday rotation.

However, the ion deflectometry does not require knowledge of the electron density distribution along the ion paths and it can be used in the vacuum environment.

The path-integrated field $\int \boldsymbol{B} \times d\boldsymbol{L}$ represents an effective sum of ion deflections along the ion path. Ion trajectories in topologically different but equivalent B-fields result in the same value of this integral. For example, suppose that an ion undergoes the same but opposite consecutive deflections or rotates by a multiple of the complete orbit. In that case, a contribution of this part of the ion trajectory to the path-integrated B-fields is zero because the velocity direction after deflections remains the same as before them ($\boldsymbol{v}_i = \boldsymbol{v}_f$). Given by the sine function in Eq. (2), there is a maximum of the measurable path-integrated B-field $||\int \boldsymbol{B} \times d\boldsymbol{L}||$, which corresponds to a half turn ($\alpha = \pi$). One orbit deflection of a 1-MeV proton means two half turns and $\sim$100 Tcm.

Because we assume only azimuthal B-fields $\boldsymbol{B}$ with the azimuthal symmetry, the components of $\int \boldsymbol{B} \times d\boldsymbol{L}$ are $\int (0, B_\varphi, 0) \times (dr, rd\varphi, dz) = (\int B_\varphi \mathrm{d}z,\ 0\ , -\int B_\varphi \mathrm{d}r)$, where d$r$ and dz are not differentials of independent spatial coordinates but components of an ion path element $\mathrm{d}\boldsymbol{L} = (\mathrm{d}r,\ \mathrm{d}z)$. Therefore, they depend on each other according to the motion equation due to the Lorentz force. A radial component of the velocity shift vector Δv_r is given by

$$\Delta v_r = v_{fr} - v_{ir} = v(\sin\Omega - \sin\theta) = -\frac{Q}{m}\int B_\varphi \mathrm{d}z. \tag{3}$$

The vectors $\boldsymbol{v}_i$ and $\boldsymbol{v}_f$ have radial components v_{ir} and v_{fr}, and make angles θ and Ω with the negative direction of the Z-axis, respectivel, as illustrated in Fig. 1. Let these angles be left-handed about the B-field vector $\boldsymbol{B}_\phi$. The angle θ is coupled with the initial velocity $\boldsymbol{v}_i$ and the ion source's divergence, so we call it the *divergence angle*. The angle Ω relates to the final velocity $\boldsymbol{v}_f$ and an incident angle at which ions are detected, so we call it the *observation angle*. We define the deflection angle α as a difference between the initial and final angles $\alpha = \Omega - \theta$. However, a sign of the divergence angle θ in Fig. 1 is opposite to the observation angle Ω, as it is set in our experimental configuration, and therefore, $|\alpha| = |\Omega| + |\theta|$. In reality, either the divergence angle θ or the observation angle Ω usually cannot be easily measured in the experiment, which complicates direct measurements of the deflection angle α and, hence, the path-integrated fields $\int \boldsymbol{B} \times d\boldsymbol{L}$. Therefore, numerical ray-tracing simulations are often necessary. By choosing the specific experimental setup, one can determine one of these angles. If the ions are emitted to the D-grid from a point, the divergence angle θ is set. If the ions are projected onto the detector through a point (e.g., a small pinhole), the observation angle Ω is established.

The recorded image in the detector displays a map of D-grid distortions (deflectometry) or simply the ion fluence (radiography), which represents the ion displacements. Because the ions are emitted along the Z-pinch axis, we record only radial displacements $\Delta r = \int \Delta v_r \mathrm{d}t$. For the path-integrated B-field measurements from the experimental data, we need to find a relation between Δr and $\int \boldsymbol{B} \times d\boldsymbol{L}$. Typically, the estimation of the path-integrated B-fields from the measured ion displacements is provided by a paraxial approximation or a specific experimental setup.

In laser-produced plasma experiments, the dimensions of the observed plasmas a are small ($\sim\mu$m) compared to their distance to the detector A ($\sim$ mm). For small proton deflections ($\sin\alpha \approx \alpha$), one can use the paraxial approximation and obtain the deflection angles $\alpha = (Q/m_p v_p)\int B\mathrm{d}a$ from the proton displacements $\Delta r \propto \alpha A$, where m_p and v_p are the proton mass and velocity, respectively. In this approximation, deflections are small deviations of the laminar proton rays emitted from the well-defined point-like source. Because the laser-produced plasmas are usually expanding and highly dynamic, their spatial scale during ion probing may be difficult, and thus, only values of the path-integrated B-field are presented. They are calculated using numerical simulations[53] and can reach up to hundreds of MG μm or a few Tcm (e.g., in Ref. 33), which correspond to deflection angles of a few degrees (100 MGμm $\sim 3°$, for 3-MeV protons).

In tokamak plasmas, the heavy ion beam probe (HIBP) diagnostic is used for the measurements of local fluctuations of the toroidal vector potential A_ϕ. It analyzes the toroidal shift $\Delta\phi$ of the probing heavy ions in the detector caused by a change in their gyroradius in a selected region of the plasma (sample volume) due to their additional ionization. The deflection angles are large because both the ion injector and the ion collector are located on the low-field side of the tokamak vessel. Again, only small fluctuations of predicted ion trajectories in the equilibrium magnetic fields are investigated.[41,42,44] The experimental data by conventional measurements of equilibrium plasma parameters are required for the numerical reconstruction of these equilibrium magnetic fields and the ion trajectories.

In the Z-pinch plasmas, we can use neither of these two approaches. The expected path-integrated magnetic fields (tens of Tcm) are too strong to use the paraxial approximation, and the adequately small ion source requires a high-intensity laser system connected to the Z-pinch device. Compared to the tokamak plasmas, we have only limited information about the topology of ion trajectories. However, in Z-pinches, one may consider only azimuthal B-fields, which are weakly dependent on a z-variable. To employ the ion deflectometry in Z-pinch plasmas, we have developed our unique experimental setup. Its three key elements are: a source of multi-MeV deuterons, investigated in our previous research,[54,55] a fiducial deflectometry grid, and, finally, a projection through a small and distant pinhole, which selects deflected deuteron beams with aligned trajectories. This experimental setup allows us to apply several approximations to estimate the path-integrated $\int B_\varphi \mathrm{d}z$ and will be discussed in Sec. III.

III. EXPERIMENTAL SETUP AND ANALYSIS TECHNIQUES

A. Experimental configuration

At present, ion sources used for ion deflectometry are usually external and well defined. For the ion beam probe diagnostics in tokamaks, ion beams are provided by an ion injector supply, which is already employed for plasma density and potential measurements. In the proton radiography of laser-generated plasmas, MeV protons are produced by the interaction of a high-intensity laser with a target[56] or a fusion capsule filled by D^3He gas.[38] In both cases, a virtual point-like source of laminar proton beams is created. Now, we introduce an alternative for the source of ions capable of probing Z-pinch B-fields. For a few past years, we have studied the acceleration of multi-MeV hydrogen ions in Z-pinch plasmas. The obtained experience enabled us to enhance both an average ion and neutron yield and better understand the ion acceleration mechanism.[54]

The experimental results discussed in this paper come from the first-ever Z-pinch-driven ion deflectometry experiments performed on the GIT-12 generator in Tomsk, Russia. The peak current of this pulsed power device is ~3.5 MA with ~1.7 μs rise time, and the circuit current during the stagnation of the Z-pinch is ~2.7 MA. The electrodes, planar meshes, are separated by the 20 − 25 mm distance. The experiments are carried out in a configuration with a hybrid gas-puff (see Refs. 54 and 57 for more details). A homogenous and uniformly conducting current sheath is created in preionized low-density (~5 μg cm^{-1}) carbon-hydrogen (CH) plasma shell injected by cable guns before the implosion. Deuterium gas load (~100 μg cm^{-1}) is puffed from supersonic nozzles and then heated and compressed by an imploding current sheath, resulting in a Z-pinch. Figure 2 illustrates a typical evolution of the imploding Z-pinch plasma. First, the plasma implodes toward the axis of the experimental setup. Afterward, a stagnation of the Z-pinch occurs, and a plasma neck is created due to plasma instabilities near the anode. The density of the perturbed plasma drops until the current is subsequently disrupted due to a rapid increase in the plasma impedance. Due to the loss of its quasineutrality, the plasma neck is depleted. Shortly after the disruption (~1 ns), a burst of the hydrogen ion beams, mostly deuterons, with energies up to tens of MeV[52] is accelerated in the current direction in the depleted gap near the anode.[54] Some protons can also be accelerated since they are present as an impurity from the preionized low-density CH-plasma shell. However, the deuterons accelerated from the dense D_2 gas-puff load predominate. During the ion emission, the azimuthal magnetic fields in the plasma near the cathode remain relatively unaffected and deflect accelerated ions arriving downstream from the depleted gap. Even during their acceleration phase, the ions are deflected by both transient electric and magnetic fields, but we assign these effects to the acceleration processes in the depleted gap and assume no ion acceleration in the plasma near the cathode in the time of ion emission. The ion beams carry information about transient fields during their *acceleration stage* near the anode, and about global B-fields during their *deflection stage* in the rest of the Z-pinch plasma. Simultaneous and unambiguous analysis of magnetic fields in both stages during one shot is impossible. Thus, we study these two stages of the ion trajectories separately. The mechanism of the efficient ion acceleration due to Z-pinch plasma neck disruption is thoroughly discussed by Klir.[54] In the present paper, we focus on the measurements of the global magnetic fields using already accelerated deuterons. Therefore, we generalize the *ion source* as the region above the D-grid. We only demand that the ion beams passing the D-grid at any point have sufficient divergence to be focused into the pinhole even in the strong B-fields.

To avoid strong deflections and complicated ion paths, the energy of the probe ions should be sufficiently high. From Eq. (2), we see that for reasonable measurements of $B_\varphi \sim 10 - 100$ T with dimensions $L \sim 1$ cm (i.e., with the path-integrated fields of $||\int \boldsymbol{B} \times d\boldsymbol{L}|| \sim 10 - 100$ Tcm), the deuteron energies ~1 − 10 MeV are favorable. Other conditions for our measurements are the sufficient fluence and divergence of the ion beams. The former is needed for a good contrast of the deflectograms; the latter is to offset the ion focusing by the azimuthal magnetic field. The spectrum of the accelerated deuterium beams in our experiments is quite broad. However, both ion divergence and fluence decrease with the ion energy, and thus, for mapping the B-field distribution, we could only utilize deuterons with energies of 1 − 5 MeV. Achieving such ion energies is, in our opinion, within reach of current MA Z-pinch devices so that similar experiments may be considered in other MA Z-pinch facilities. In cooperation with the team working in Naval Research Laboratory in Washington, DC, on the HAWK generator, we have accelerated deuterons up to energies >7 MeV[55] with sufficient ion fluence. Moreover, ~0.1 − 5 MeV protons have been reported on the MAGPIE pulsed-power generator.[58]

Conventionally, the ion deflections are manifested using a fiducial mesh located between a source of the ion backlighting and the region of interest. However, in our case, the ion source is internal, that is, located together with the studied magnetic fields inside the Z-pinch plasma. To separate the source and the B-fields and to measure the ion deflections in our experiments, we place a circular stainless-steel (SS) deflectometry grid (*D-grid*) in the anode–cathode gap (see Fig. 3). The diameter of the D-grid is 20 mm. The D-grid has 2-mm-wide square openings and 0.5-mm-thick wires, and it is attached to the cathode mesh by a 10-mm-high and 330-μm-thick stainless-steel stalk wire. Figure 4 shows a series of SXR images obtained by a multichannel plate (MCP) camera, which captures the evolution of the imploding plasma with the inserted D-grid. As seen, the plasma splits and propagates above and below the D-grid, effectively creating two separated and partly independent Z-pinches. The plasma below the D-grid is delayed by several ns, and so, at the time of the current disruption and the ion emission, it is still imploding toward the axis. Surprisingly, in these experiments with the inserted D-grid, we detected ion beams with very high energies, reaching up to at least 55 MeV.[52] The height of the D-grid is an important parameter for the ion acceleration mechanism. If the D-grid is put too close to the anode, it disrupts the acceleration mechanism and the ion emission. On the other hand, if the D-grid is put too close to the cathode, the ion acceleration is not

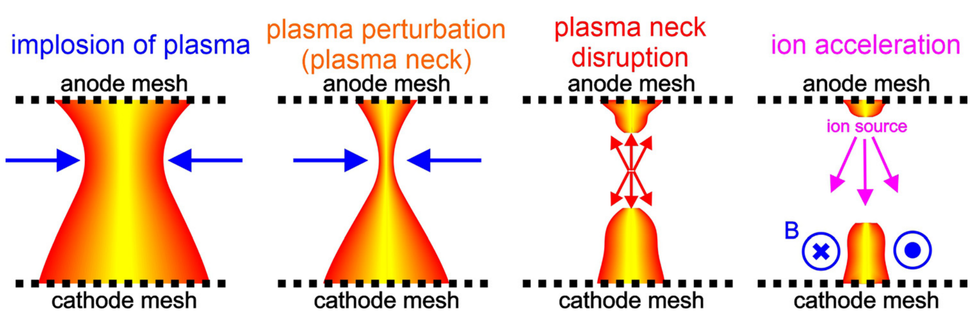

FIG. 2. Cartoon of a typical evolution of the Z-pinch plasma on GIT-12: plasma implodes toward the axis between the electrode meshes due to $\boldsymbol{J} \times \boldsymbol{B}$ force. A strangulated necked plasma appears near the anode due to the plasma instabilities, until it is disrupted. Shortly after, hydrogen ions (mostly deuterons) are accelerated in the depleted region after disrupted necked plasma and are deflected in azimuthal magnetic fields in the remaining plasma, which stay relatively unaffected by the disruption.

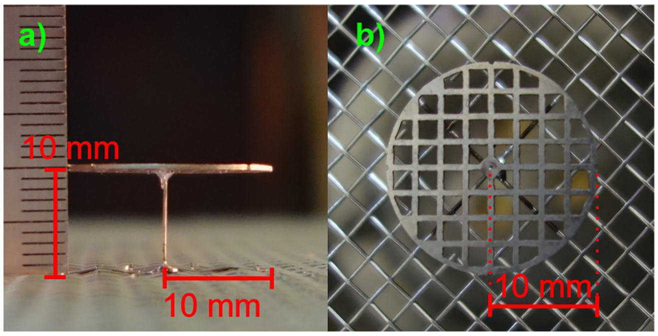

FIG. 3. Side view **(a)** and top view **(b)** of a circular stainless-steel (SS) deflectometry grid (D-grid). Its diameter is 20 mm, and it is connected by the 10-mm-high and 330-μm-thick stalk SS wire to the cathode mesh.

affected, but the current remains at the D-grid edge and does not penetrate below it. The D-grid does influence the plasma imploding but only to the extent that it separates the ion source from the rest of the plasma. The stalk wire supporting the D-grid may glow but carries only a small portion of the total current (see Sec. IV C). Because it prevents the depletion of the plasma on-axis, ions cannot be accelerated below the D-grid, which would complicate the analysis of the deflectogram and the B-field measurements.

B. Deflectometry setup

The setup of the ion deflectometry diagnostics is shown in Fig. 5. Probe deuterons are emitted downstream from the extensive and divergent source near the anode and intersect the D-grid plane at the initial radial positions r_0. Here, their divergence is given by angle θ. A shadow of the D-grid is imprinted into the passing ions, and then, it becomes distorted when the ions are deflected in the azimuthal magnetic fields below the D-grid. When the ions reach the cathode plane, their radial positions are r_C. Because free electrons effectively neutralize a net charge of the ion beams in the plasma and the B-field are assumed negligible below the cathode mesh, the ions continue along the straight lines toward the pinhole camera, placed on the axis below the cathode mesh. The pinhole distance from the cathode is $l_{pin} = 105$ mm, and it determines the observation angle of the ions Ω at the radius r_C.

The pinhole camera is a metal container with one or a few tiny pinholes ($250 - 500\,\mu$m in a diameter) bored in its top. The spacings between the pinholes are small compared to the distance between the cathode and the ion camera. Thus, we approximate these pinholes by only one pinhole located in the center of the setup. Inside the camera, there is a stack of radiochromic films (RCF) with Al filters. The ions are projected through a pinhole onto the detector stack and cause a measurable change of color in RCF, indicating a fluence of the detected ions. Here, the final image (deflectogram) of the distorted D-grid shadow is finally recorded. Distortions of the deflectogram created by ion displacements Δr are proportional to $\int B_\varphi dz$. Each position of RCF detectors in the stack is coupled to a Bragg peak of the ion stopping power. We can estimate the lower energy threshold for the ion signal in each layer using Stopping and Range of Ions in Matter[59] (SRIM) simulations.

Suppose there would be no fields [see Fig. 5(a)], then, only ions directing right into the pinhole would be detected and the deflectogram would show a scaled original image of the undistorted D-grid shadow. We can use the corresponding ion positions in the cathode plane r'_0 as a reference for measuring the ion displacement Δr. We measure these displacements at the cathode because the experimental deflectograms show scaled images of the deflected ion beams in the cathode plane r_C due to the direct projection through the point-like pinhole between the cathode and the detector plane.

In the conventional setup, a fixed point of the virtual source is used to set the divergence angle θ. In contrast, we have the extensive ion source, but we use a fixed point of the pinhole to set the observation angle Ω [cf. Fig. 5(b)]. Due to the large cathode-pinhole distance l_{pin} compared to the ion radial position r_C at the cathode, the pinhole camera captures the ions at a very small observation angle ($\Omega \lesssim 7°$). Then, the divergence angle θ is nearly equal and opposite to the deflection angle α. It means that, in strongly focusing magnetic fields, ion beams leaving the D-grid must be highly divergent ("defocused") to be detected after their deflection. In other words, the source must emit ions far outwards from the axis, so they can be focused toward the axis into the pinhole by the azimuthal B-fields. If $\Omega \approx 0$, we can simplify Eq. (3) to

$$|\sin\theta| \approx |\sin\alpha| = \frac{Q}{mv}\int B_\varphi dz = \int \frac{dz}{R_L}, \tag{4}$$

where R_L is an ion Larmor gyroradius. For each ion trajectory, we can study only the radial component of the path-integrated field

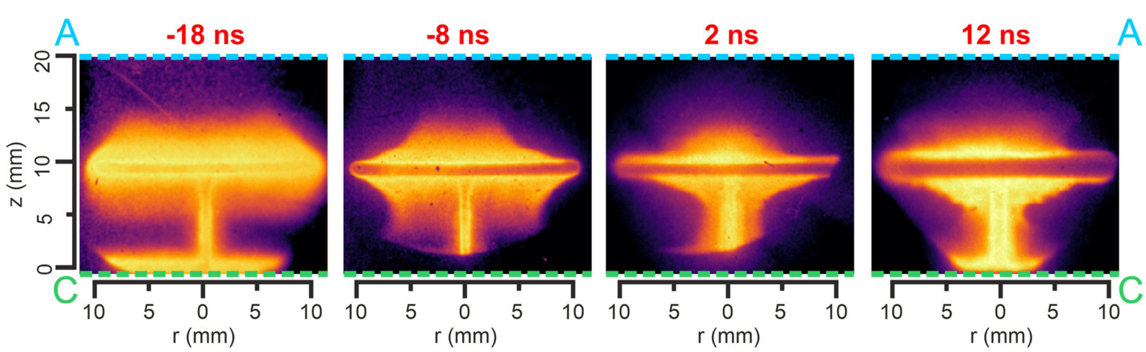

FIG. 4. SXR images of the evolution of the imploding plasma with the inserted deflectometry grid obtained by a multichannel plate (MCP) camera. The $t = 0$ ns corresponds to the onset of >2 MeV gamma, neutron, and ion emission.

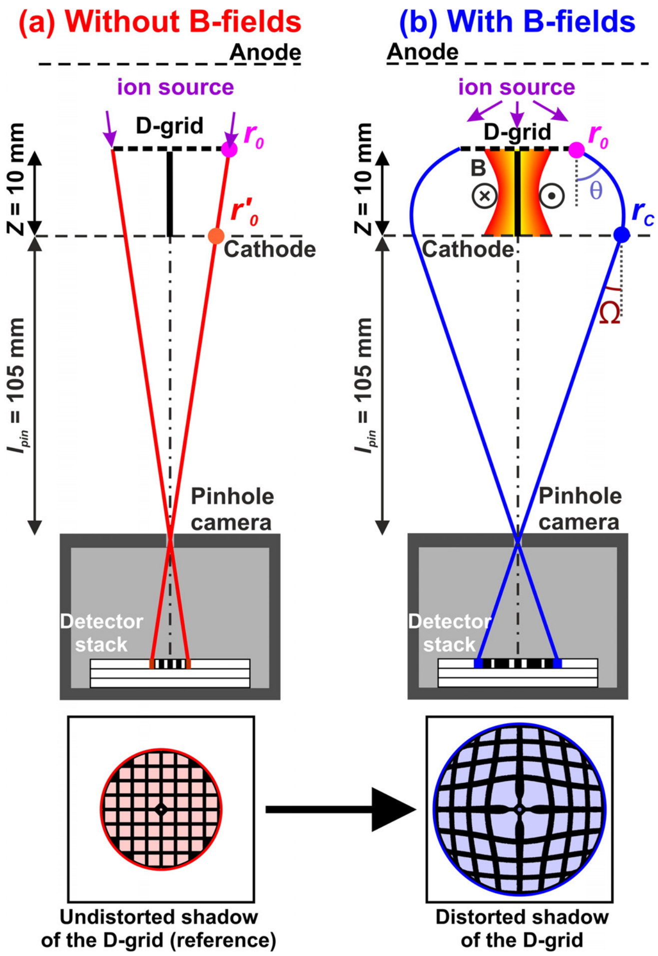

FIG. 5. Not-to-scale scheme of the deflectometry setup. Ions are accelerated downstream from the extensive and divergent source in the region above the D-grid. A pattern of the D-grid is imprinted into the passing beams and recorded by a distant pinhole camera. In the case of no fields below the D-grid **(a)**, the ions must aim directly from the D-grid into the pinhole to be detected. The D-grid shadow is not distorted and shows us a reference image. In the case of non-zero azimuthal B-fields below the D-grid **(b)**, the ions must leave the D-grid at a certain divergence angle θ because the focusing B-fields deflect them toward the axis and into the pinhole. Due to ion deflections, the D-grid shadow becomes expanded and distorted.

$(\int \boldsymbol{B} \times d\boldsymbol{L})_r = \int B_\varphi \mathrm{d}z$ instead of its strength $||\int \boldsymbol{B} \times \mathrm{d}\boldsymbol{L}||$. This approximation holds as long as $\sin(\alpha) \approx 2\sin(\alpha/2)$ [cf. Eqs. (2) and (4)], which can be applied to moderate ion deflections by deflection angles $\alpha < 50°$. For every point in the measured distribution of a *deflection field* (a map of D-grid displacements in the deflectogram), we can find a corresponding value of an averaged B-field $\overline{B_\varphi} = \int B_\varphi \mathrm{d}z/Z$, where Z is the D-grid-cathode distance. Because we maximize the axial scale of the studied B-field region by $Z = 10$ mm, the averaged B-field is the low estimate of the actual B-fields, even if perturbed (see Sec. IV D). If deflection angles are not very small and one cannot use the paraxial approximation, the averaged B-fields $\overline{B_\varphi}$ are usually estimated by reproducing the experimental data via numerical ray-tracing simulations. However, in our experiments, we can approximate the ion trajectories by Larmor orbits and estimate $\overline{B_\varphi}$ analytically using a *Larmor orbit approximation*.

C. Larmor orbit approximation

If the B-field strengths are moderate and the gradients are gentle, a curvature of an ion trajectory changes slowly along the ion path. In that case, we can approximate this ion trajectory by a Larmor orbit with an averaged radius $\overline{R_L}$ [see Fig. 6(a) and 6(b)] and calculate an averaged Larmor B-field $\overline{B_L}$.

The deflectometry setup provides enough information to find $\overline{R_L}$ for each ion trajectory from the measured quantities [see Fig. 6(a)]: the initial ion position r_0 at the D-grid, the final position of the deflected ion r_C at the cathode, and a tangent line of the ion trajectory at the cathode determined by the observation angle Ω. Because $\Omega \approx 0$, the initial radial ion position r_0 is almost equal to the reference ion position r'_0 at the cathode ($r_0 - r'_0 < 1$ mm). Therefore, we define the ion displacement in the cathode plane as $\Delta r \approx r_C - r_0$. By solving a simple geometric problem, we obtain a relation for Larmor B-fields $\overline{B_L}$

$$\overline{B_L} = \frac{mv}{Q} \frac{2l_{pin}\Delta r + 2Z(\Delta r + r_0)}{\left[(\Delta r)^2 + Z^2\right]\sqrt{(\Delta r + r_0)^2 + l_{pin}^2}}. \tag{5}$$

Figure 6(c) shows that $\overline{B_L}$ calculated by Eq. (5) is almost independent to the initial radial ion position r_0. Moreover, the curves shown in Fig. 6(c) have a single maximum, which roughly corresponds to the maximum of the sine in Eq. (4), when the deflection angle $\alpha = \pi/2$ and $\Delta r \approx \overline{R_L} \approx Z$. For weaker (moderate) B-fields ($\overline{B_L} \lesssim 30$ T for 2.3-MeV deuterons), $\overline{B_L}$ grows linearly with increasing ion displacements Δr at the cathode. Because we assume that the Larmor orbits approximate the ion trajectories, then $\overline{B_L}$ approximates the averaged B-field $\overline{B_\varphi} = \int B_\varphi \mathrm{d}z/Z$. Then, the linear part of the curves in Fig. 6(c) can be fitted by

$$\Delta r \approx \frac{Q}{2mv}\overline{B_L}Z^2 = \frac{Q}{2mv}Z\int B_\varphi \mathrm{d}z. \tag{6}$$

Therefore, we obtain a simple and linear relation between the ion displacements $\Delta r = \int \Delta v_r \mathrm{d}t \propto \int\int B_\varphi \mathrm{d}z\mathrm{d}t$ and the Larmor B-fields $\overline{B_L}$. We will demonstrate the Larmor orbit approximation on an example of the B-field profile [see Figs. 6(b) and 6(d)]. The azimuthal B-fields are axially uniform and linearly increasing with a radius up to R_p = 9 mm, where the total pinch current becomes 1.5 MA. The choice of the 9-mm-radius Z-pinch allows us to study B-fields both inside and outside the Z-pinch. In these simulations, a deuteron source is divergent, extensive (broad region above the D-grid), and monoenergetic (2.3 MeV).

To retrieve the averaged B-fields from the deflectograms, we need to assign the estimated B-fields $\overline{B_\varphi}$ to some radial position between r_0 and r_C. It is reasonable to choose an averaged radial position $\bar{r}$ integrated along the ion trajectory by the axial ion position z, that is, $\bar{r} = \int r\mathrm{d}z/Z$, which characterizes the mean radius of the ion trajectory. In Fig. 6(b), our ion-tracing simulations show that for moderate deflections, this averaged radius $\bar{r}$ is close to the measured radial position of the ions in the cathode plane ($\bar{r} \approx r_C$). Because the observation angle Ω of the detector is set low by the large pinhole-cathode distance l_{pin}, we detect only deflected ion beams that are aligned almost parallelly to the Z axis by the focusing B-field during their deflections and the ion trajectories have similar shape near the cathode.

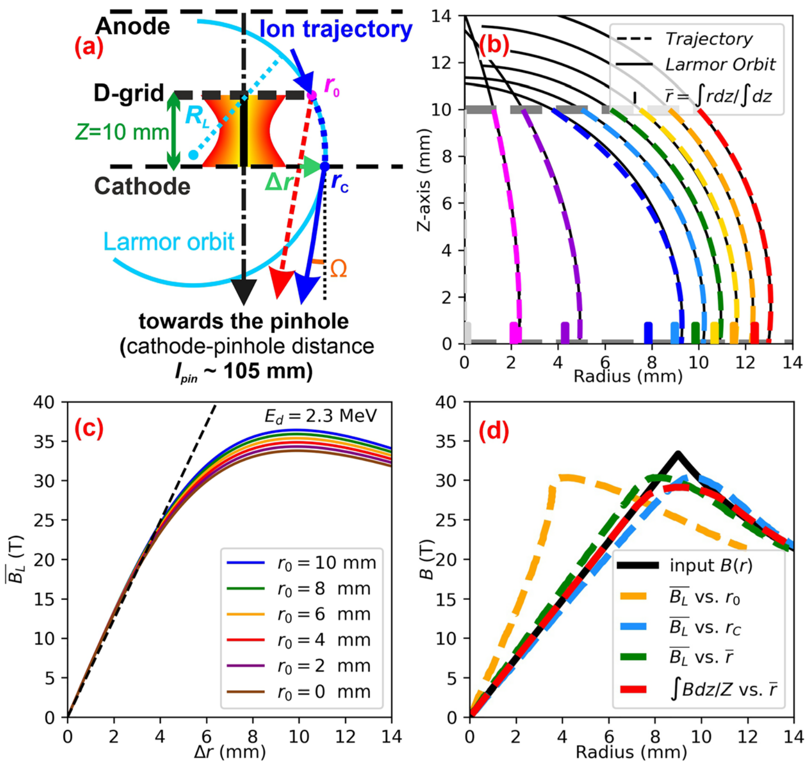

FIG. 6. (a) Moderate ion deflections can be approximated by a Larmor orbit approximation, which assumes small variations of the B-fields along the ion path. **(b)** Simulated trajectories of ions, deflected in azimuthal B-fields, can be quite well approximated by Larmor orbits. In this example, the synthetic B-field is axially uniform and grows linearly ($B_\varphi(r) \propto r$) up to 9 mm radius, where the pinch current reaches $I = 1.5$ MA. These parameters are used for demonstration purposes of this approximation. **(c)** The Larmor-orbit B-field $\overline{B_L}(\Delta r, r_0)$ given by Eq. (5) depends weakly to the initial ion position r_0 at the D-grid. For moderate strengths, $\overline{B_L}$ is linear with respect to the radial displacements Δr according to Eq. (6). **(d)** Reconstruction of the radial B-field profile $B_\varphi(r)$ (black lines), which is the same as in a figure (b). First, it is reconstructed from the measured displacements using the Larmor orbit approximation (yellow, blue, green lines), where $\overline{B_L}$ are assigned to the ion positions at the D-grid r_0, at the cathode r_C, and then, to the averaged radius $\bar{r} = \int r dz/Z$, respectively, and finally by numerical ion-tracking simulations (red line), where $\int B_\varphi dz/Z$ are assigned to averaged radius $\bar{r}$. The maximum of this B-field distribution is 33 T, which is slightly beyond the limit of the moderate B-fields for 2.3-MeV deuterons.

Therefore, the radii near r_C have greater weights in the integral $\int r dz$. Figure 6(d) illustrates how well the ion deflections can reproduce the radial profile of B-fields implemented in the simulations.

For B-fields up to $B \lesssim 30$ T, the deflections of the 2.3-MeV deuterons are moderate, and the averaged B-fields $\overline{B_\varphi} = \int Bdz/Z$ copy the input B-field very well. The best fit is if $\overline{B_\varphi}$ is assigned to the averaged radius $\bar{r}$ (red curve). However, even if $\overline{B_\varphi}$ is assigned to the radial ion position at the cathode r_C (blue curve), the fit is still good.

For $B > 30$ T, the deuteron deflections are too strong, and ions lose the ability to measure B-field locally. Due to our choice of the current 1.5 MA at the 9 mm radius (∼33 T) in our example, we show the performance of the ion deflectometry slightly beyond the B-field 30-T limit of the Larmor approximation for 2.3 MeV deuterons. Since the shape of the ion trajectories is set by Eq. (4), the divergence angle θ must grow with stronger B-fields and, hence, larger deflection angles α. Therefore, the ion trajectory becomes more curved close to the anode (near the initial radius r_0). Then, the distance between $\bar{r}$ and r_C becomes relatively large [see blue indicator in Fig. 6(b)] and ion deflections stop reflecting the B-field distribution [see Fig. 6(d)]. Similar problems appear if the B-fields gradients are too steep. Comparing the red curve of simulated averaged B-fields $\overline{B_\varphi}$ to the Larmor B-fields in Fig. 6(d), we observe that the retrieved B-field profile smoothing does not originate from the Larmor orbit approximation. On the contrary, it shows the inherent integrating behavior of ion deflectometry if the B-field magnitudes or gradients are too strong.

In Subsection III D, we present typical patterns of synthetic deflectograms and reconstruct their corresponding radial distributions of the axially uniform B-fields within the limits of moderate deuteron deflections.

D. Analysis of the synthetic deflectogram patterns for characteristic radial profiles of azimuthal B-fields

The D-grid displacements in the deflectograms manifest the radial distribution of ion deflections and path-integrated B-fields. To understand the experimental deflectograms and interpret their structure, we present synthetic deflectograms generated by our ion-tracking numerical code using characteristic radial distributions of axially uniform azimuthal B-fields $B_\varphi(r)$. The experimental setup is implemented in the simulations. The pinch radius is $R_p = 9$ mm, where the total pinch current reaches 1 MA. We use 2.3 MeV deuterons, and

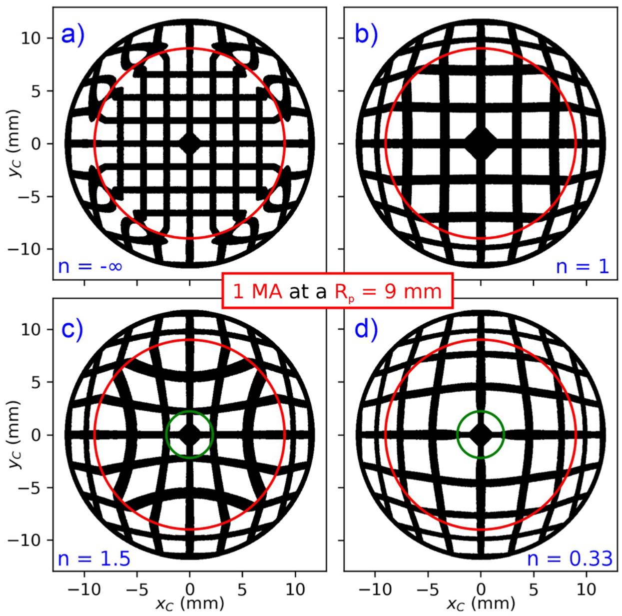

FIG. 7. Demonstration of typical synthetic deflectograms for four radial distribution of the B-field $B_\varphi(r)$ inside the pinch with a radius $R_p = 9\,\text{mm}$ (red circle) where total pinch current reaches 1 MA. In the area within the pinch radius R_p, the B-fields are given by $B_\varphi(r < R_p) = \mu_0 I r^n / 2\pi R_p^{n+1}$. Four typical profiles of current densities $j_z(r < R_p) \propto n r^{n-1}$ are coupled to four values of the exponent n: **(a)** $n = -\infty$: no current density inside the Z-pinch [$B_\varphi(r < R_p) = 0$]; **(b)** $n = 1$: constant current density; **(c)** $n > 1$: radially increasing current density; **(d)** $0 < n < 1$: radially decreasing current density. In the last two cases, there is no B-field inside the radius $R_{cs} = 2.2\,\text{mm}$ (green circle). Outside the pinch volume (i.e., for $r > R_p$), there is no current density and so $B_\varphi(r \geq R_p) = \mu_0 I / 2\pi r$. The images are scaled to the corresponding size at the cathode plane.

thus, the maximum B-field strength of $\sim$22 T lies in the linear part of the curve in Fig. 6(c).

For a demonstration of the typical distortions of the D-grid shadow in the deflectograms, we present four synthetic deflectograms in Fig. 7. In the region within the pinch radius ($r < R_p$), delimited by a red circle, we compare four characteristic radial distributions of azimuthal magnetic fields $B_\varphi(r < R_p) = \mu_0 I r^n / 2\pi R_p^{n+1}$ with current densities $j_z(r < R_p) \propto n r^{n-1}$ according to four cases of the exponent n:

(a) $n = -\infty$: a thin skin-current sheath with no B-field inside; **(b)** $n = 1$: uniform current density; **(c)** $n > 1$: increasing current density with a radius; **(d)** $0 < n < 1$: decreasing current density with a radius. In the two last cases, the current sheath is shifted and there is no B-field within the radius $R_{cs} = 2.2\,\text{mm}$ (green circle), which roughly agrees with the experimental data presented in Subsection IV C. Outside the pinch ($r > R_p$), there is zero current density and all B-field distributions are identically $B_\varphi(r \geq R_p) = \mu_0 I / 2\pi r$.

For $r > R_p$, the distorted D-grid pattern has the same shape and structure for all examples, with the only exception of **(a)**. In the case of the skin current, the B-field gradient is too steep, so the deflecting ions cannot quickly change their trajectories and accurately copy the jump in the B-field profile. Thus, the deflected ions create a broader distorted transition in the deflectogram near the Z-pinch radius of R_p.

For $r < R_p$, the D-grid shadows are substantially distinct. In general, the ion deflections cause an expansion of the D-grid shadow. The D-grid pattern structure in the deflectogram **(b)** keeps the squared but magnified shape and resembles the deflectogram with no B-field **(a)**. The deflectogram **(c)** shows an increasing trend of ion displacements with an increasing radius. Because the D-grid pattern is more displaced in the corners of the square openings, the deflected ions create a concave pattern (*pincushion distortion*). In contrast, the ion displacements in the deflectogram **(d)** are decreasing with an increasing radius. The square D-grid openings are "*inflated*" outward, which results in a convex pattern (*barrel distortion*) of the D-grid shadow. Moreover, notice that in the deflectogram **(d)**, the transition of the distortions near the radius R_p is very smooth.

These observations can be explained when we compare ion displacements Δr and the implemented distributions of the B-fields $B_\varphi(r)$ (see Fig. 8). As seen, the distributions of the ion displacements in the cathode plane Δr follow the radial profile of the path-integrated B-fields $\int B_\varphi \mathrm{d}z$, according to Eq. (6). Because we input the axially uniform B-fields, the averaged B-fields $\overline{B_\varphi} = \int B_\varphi \mathrm{d}z / Z$ approximate these B-fields B_φ. Due to the linear relation between Δr and B_φ, the derivative of the path-integrated B-fields and the distribution of the radial current density in the cylindrical coordinates $j_z(r < R_p) = \partial(r B_\varphi)/(r \partial r)/\mu_0$ are proportional to displacement gradients $\partial(r_C \Delta r)/(r_C \partial r_C)$, which characterize the deflectogram structure. In both examples **(a)** and **(b)**, the current density $j_z(r < R_p)$ is constant (in the former case, zero) and creates the regular squared grid pattern. In the other deflectograms **(c)** and **(d)**, the current density profile $j_z(r)$ also dictates the distortion gradients. Near the pinch radius R_p in the B-field profle **(d)**, the current density decreases so low that the pattern of the deflectogram within the red circle is similar to the outer region where j_z falls to zero, and hence, the transition in distortion at $r = R_p$ in Fig. 7(d) is hardly noticeable.

In conclusion, for moderate ion deflections and gradients of azimuthal B-fields, the magnitude of $\Delta r(r_C)$ is proportional to $\int B_\varphi \mathrm{d}z(r_C)$. If the B-fields are axially uniform, $\partial(r_C \Delta r)/(r_C \partial r_C)$ and

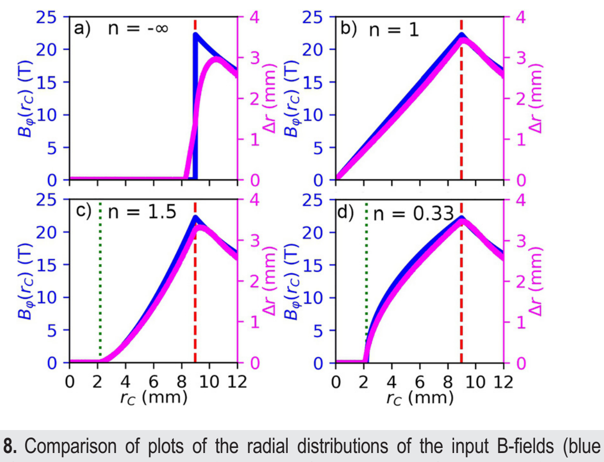

FIG. 8. Comparison of plots of the radial distributions of the input B-fields (blue curves) and displacements Δr of the D-grid (magenta curves), measured from the synthetic deflectograms in Fig. 7, according to the radial ion position in the cathode plane r_C. The total pinch current of 1 MA flows within the pinch radius $R_p = 9\,\text{mm}$ (red dashed line). In the deflectograms **(c)** and **(d)**, we set no B-field within a radius $R_{cs} = 2.2\,\text{mm}$ (green dotted line).

the structure of the deflectogram (barrel, pincushion, etc.) are coupled to the current density profile j_z, mapped by the deflecting ions.

IV. EXPERIMENTAL RESULTS SUPPORTED BY OUR NUMERICAL SIMULATIONS

A. Experimental deflectograms

In Fig. 9, we present selected experimental deflectograms with distorted D-grid shadows and compare them to reference ("undistorted") photographs of the original D-grids in the shots Nos. 2402, 2404, 2408, 2418, and 2420. The images are scaled to their corresponding size in the cathode plane. These are first-ever experimental data obtained via the Z-pinch-driven ion deflectometry and the first deflectograms of the Z-pinch using axially emitted ions. For better clarity, the deflectograms are presented in false colors with enhanced contrast. Parts of the detector irradiated by ions are bright, and the D-grid shadows are dark. The ions must traverse both the D-grid and the cathode mesh (cf. Fig. 5), and thus, shadows of them both are evident in the deflectograms. The meshes are rotated by the 45° angle relative to each other for a better distinction between them. Partially cut out the cathode mesh center helps a better, more unobstructed view of the D-grid center. The cathode-mesh shadow is not distorted because, below the cathode, ion trajectories are undeflected. Ion backlighting in the images is not uniform. The absence of the ion signal in the deflectograms is caused by either a nonuniform ion source or ion beams deflected somewhere outside the pinhole (i.e., by not satisfying projection conditions).

Nevertheless, some parts or at least the edge of the distorted D-grid shadow are apparent on all shots in Fig. 9.

In the deflectogram of the shot No. 2402, the size and shape of the D-grid shadow are nearly identical to its original undistorted image. In this shot, the D-grid was placed closer to the cathode (~6 mm), which apparently prevented the current sheath below the D-grid from penetrating closer to the axis. Only a slight distortion near the D-grid edge indicates nonzero magnetic fields. In the other shots, the D-grid was located in ~10 mm distance from the cathode mesh. The current sheath propagated closer to the axis, and these deflectograms show the deformed D-grid shadows.

The deflectograms provide different amount of information according to their quality due to their clarity, which vary shot-to-shot due to the imperfect backlighting or blurry features of the D-grid shadow. In some shots, the D-grid shadow is not clear enough for retrieval of a B-field distribution. The expansion of the D-grid edge caused by the ion deflections allow us to estimate the total current flowing within reach of ion deflections, and it will be discussed in Sec. IV B.

Because the deflectograms show time-integrated images of ion beams deflected in the imploding plasma below the D-grid, the studied B-fields could vary during the ion probing ($\lesssim 10$ ns). According to the maximum implosion speed of the plasma, $\sim 4 \times 10^5$ m/s, the current sheath can move by ~4 mm during the 10-ns ion emission. However, if it were true, then the D-grid shadow would completely disappear. Although it might partly contribute to the low quality of some deflectograms (demonstrated by the shot No. 2418 in Fig. 9, for example), it is not a severe problem if the D-grid shadow is visible. Therefore, the emission of ions with the specific energy is much shorter or the plasma does not move much during this period. Moreover, we do not even observe multiple shadows or significant discontinuities of the D-grid caused by multiple exposures. On the other

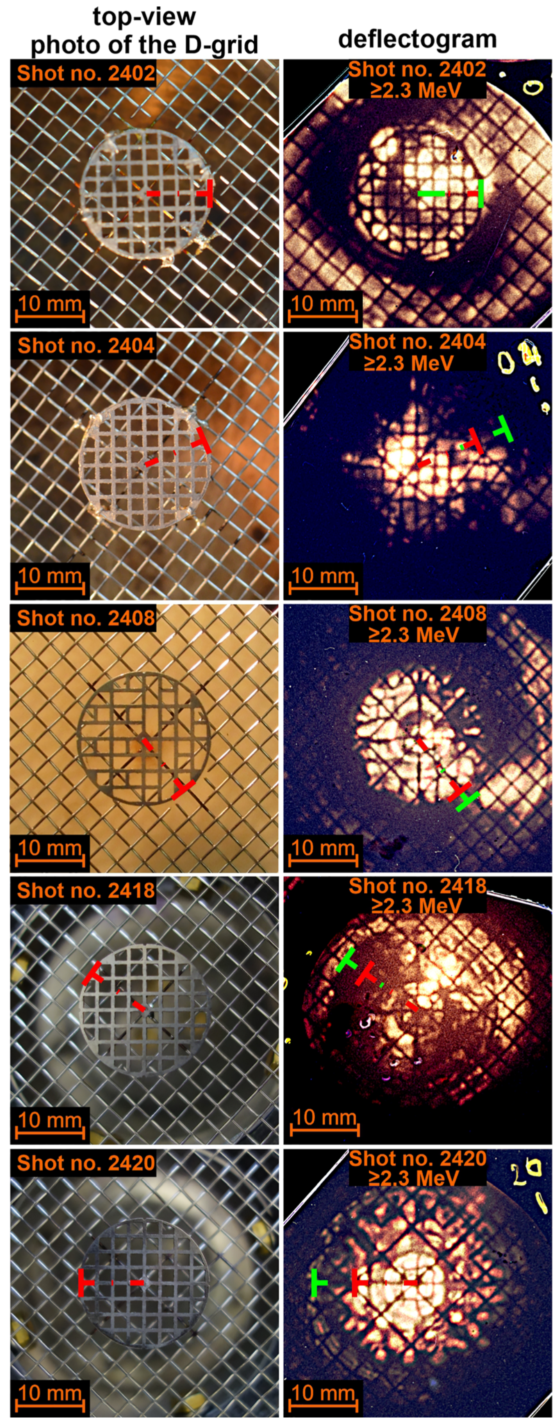

FIG. 9. Comparison of top-view photographs of the D-grid connected to the cathode mesh and the experimental deflectograms obtained in shot Nos. 2402, 2404, 2408, 2418, and 2420. Bright colors correspond to high ion fluence. The enlargement and distortion of the D-grid in the deflectogram are caused by ion deflections proportional to measured magnetic fields. Red and lime lines show the initial and displaced radius of the D-grid edge. The energy of the detected deuterons is ≥ 2.3 MeV. The images are scaled to their corresponding size in the cathode plane.

hand, the low image contrast may originate in ion scattering at the obstacle edges (i.e., D-grid or pinhole) or by the pinhole-camera resolution ($\delta \sim 0.8$ mm). However, larger complications arise from the deuteron energy estimation. All the experimental data discussed in this paper are recorded in the first RCF layers in the stack. They are shielded by a 30-μm-thick Al layer, and hence, their signals correspond to the deuteron energy threshold of 2.3 MeV. The detection probability of ions with higher energies is lower but not negligible. The second detector layer in the stack corresponds to 4.7-MeV deuterons, and thus, there is a relatively wide interval of possible deuteron energies. The beam energy estimation is the biggest but removable source of uncertainty in our measurements, and it will be addressed in future campaigns. Since the ion deflections grow with $\alpha \propto 1/\sqrt{E}$ according to Eq. (2), in theory, the path-integrated B-fields might be larger by a coefficient $\sqrt{4.7\text{-MeV}/2.3} \simeq 1.43$, that is, roughly 40%. Nevertheless, we will assume that most of the signal is created by monoenergetic 2.3-MeV deuterium beams but consider our results as low estimates of the measured quantities.

B. Estimation of the pinch current from the displacement of the circular D-grid edge

Displacements of the D-grid edge shadow provide undemanding measurements even from low-quality deflectograms, where the pattern of the distorted D-grid shadow is unclear or missing (see deflectogram of the shot No. 2418 in Fig. 9). Without the D-grid pattern, we have no information about distributions of the path-integrated B-fields, but we can estimate the value of the pinch current flowing within reach of ion deflections. To analyze the effects of the B-field distribution on the current measurements, we compare the results of several possible distributions in our ion-tracking simulations in Fig. 10.

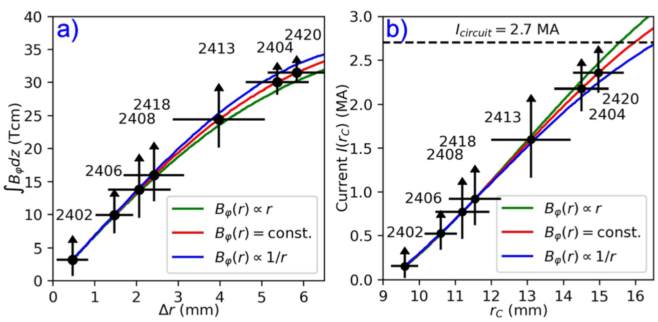

FIG. 10. (a) Estimation of $\overline{B_\varphi}Z$ using the Larmor approximation [Eq. (5)] from the measured displacements Δr of the D-grid edge in the deflectograms of the selected shots (black dots). Plots of the simulated path-integrated fields $\int B_\varphi dz$ for three radial distributions of axially uniform azimuthal B-fields $B_\varphi(r)$ according to the edge displacements are close to each other. Therefore, from the single-point measurements of the averaged radius of the D-grid edge in the deflectograms, we can only estimate a single value of path-integrated or averaged B-fields $\overline{B_\varphi}$. **(b)** Calculated averaged enclosed current $\bar{I}(r_C) = \overline{B_\varphi}2\pi r_C/\mu_0$ (black dots) and simulated currents $I(r_C)$ (lines) flowing within the radius r_C for these three B-field profiles are similar. The results show most of the 2.7-MA stagnation current flowing within the $\sim$15 mm distance from the axis during ion probing in the shots Nos. 2404 and 2420. Error bars in the measured points are given by uncertainty of estimating the averaged radial displacement of the D-grid edge in the selected deflectograms. Arrows from these points show that these results are low estimates of the current.

We study three axially uniform azimuthal B-fields $B_\varphi(r)$ with three different radial profiles: linear B-fields $[B_\varphi(r) \propto r]$, uniform B-fields $[B_\varphi(r) = \text{const.}]$, and decreasing B-fields $[B_\varphi(r) \propto 1/r]$. The first B-field profile corresponds to a constant current density, the second to a decreasing current density with radius as $1/r$, and the last B-field profile is outside the Z-pinch. In these simulations, an extensive and divergent source of monoenergetic 2.3 MeV deuterons backlights the D-grid. Figure 10(a) shows plots of simulated path-integrated B-fields $\int B_\varphi dz$ for these three distributions according to a radial shift Δr of the D-grid edge shadow in the cathode plane. The plots are very close to the uniform B-field curve, which corresponds to the Larmor orbit approximation. Since we utilize the displacement of a single deflection point (D-grid edge) from each experimental deflectogram, the measurements are rather insensitive to the radial distribution of $\int B_\varphi dz$ and the axially uniform B-fields, and follow the relation in Eq. (5). Therefore, we can use the Larmor approximation for the evaluation of path-integrated B-fields $\overline{B_\varphi}Z$ in the selected shots [black dots in Fig. 10(a)]. We consider the symmetric azimuthal displacement Δr of the averaged radius of the D-grid edge, highlighted in Fig. 9. For shot Nos. 2404 and 2420, the measured path-integrated B-fields reach up to $\sim$31.5 Tcm, which is at the limit of the Larmor approximation for the given deuteron energy. In addition, we compare the currents $I(r_C)$ flowing within the reach of deflected ions, in Fig. 10(b). Again, the curves are close to the uniform B-field curve, which shows that the Larmor orbit approximation is a useful estimation for our measurements. In shot Nos. 2404 and 2420, the estimated averaged enclosed currents $\bar{I}(r_C) = \overline{B_\varphi}2\pi r_C/\mu_0$ reached up to $\sim$2.4 MA at the $\sim$15-mm radius, which represents the majority of the total 2.7 MA circuit current during the stagnation. Error bars in the measured points are given by uncertainty of estimating the averaged radial displacement of the D-grid edge in the selected deflectograms. The maximal deviation error of the D-grid edge localization can be $\sim$1.1 mm, which corresponds to maximal uncertainty of $\sim$20%, according to Eq. (5). However, these values describe only the statistical errors that are still much smaller than the uncertainty of the beam energy determination ($\sim$40%), which can lead to higher B-fields and currents. Therefore, the arrows in Fig. 10 remind us that these results are low estimates.

In Subsections IV C and IV D, we will investigate deflectograms, which provided information about the displacement field of many points representing the distorted D-grid pattern, which manifests the radial distribution of the path-integrated B-fields.

C. Mapping (x-y) profile of the azimuthal B-fields using the distorted D-grid pattern

If the deflectograms reveal a substantial part of the D-grid shadow, we can measure (x-y) distributions of the ion displacements and path-integrated B-fields. Now, we focus on the experimental data in the featured shots with numbers 2404, 2408, and 2420 [see Fig. 11(a)]. Highlighted parts of the distorted D-grid shadow in Fig. 11(b) correspond to representative points of the original D-grid, that is, nodes and midpoints of the D-grid pattern, which are recognizable in the deflectograms. Displacements of these points represent averaged ion deflections in $\sim$1.25-mm-long and $\sim$0.5-mm-wide D-grid segments, as if these ions would go through the D-grid structure. From the measured positions and displacements of these points, we establish a displacement field [black arrows in Fig. 11(c)], to which we assign averaged B-fields $\overline{B_\varphi} = \int B_\varphi dz/Z$ using the Larmor approximation.

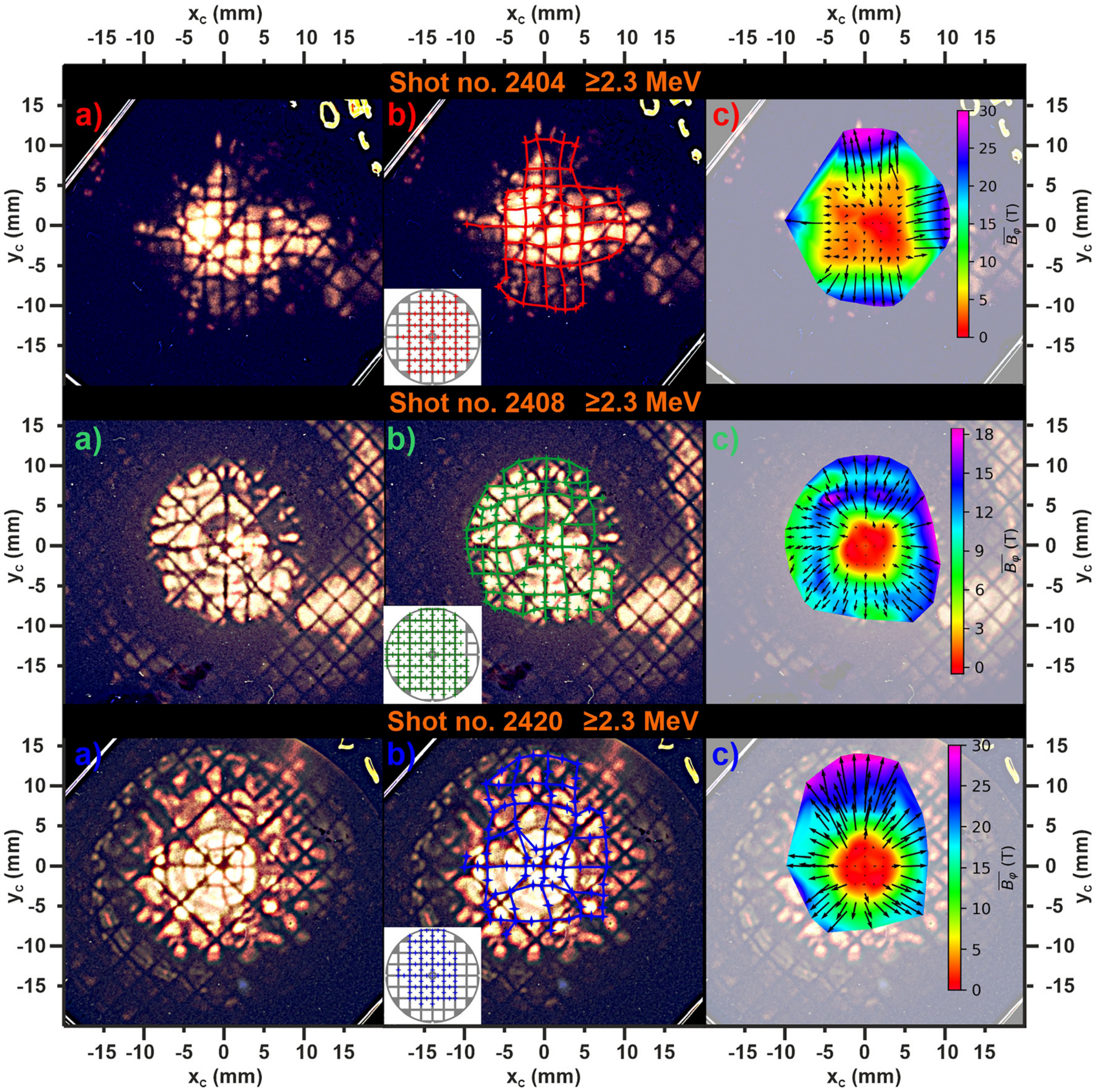

FIG. 11. (a) Original experimental deflectograms in shot Nos. 2404, 2408, and 2420 showing shadows of both the distorted D-grid and the undistorted cathode; **(b)** Deflectograms with retrieved representative points of the D-grid. Corresponding sections of the original D-grid image are highlighted in the insets. **(c)** The (x-y) maps of the averaged azimuthal B-fields $\overline{B_\varphi} = \int B_\varphi dz/Z$, calculated via the Larmor orbit approximation. The black arrows indicate a field of measured displacements $(\Delta x, \Delta y)$ of the retrieved D-grid. Spatial scales of all images correspond to the plane of the cathode mesh.

Via a cubic interpolation, we obtain 2D (x-y) maps of $\overline{B_\varphi}(x_C, y_C)$ for all three selected shots [color maps in Fig. 11(c)]. The values at the edge of the maps correlate with the results in Fig. 10. In the shot Nos. 2404 and 2408, we observe a significant anisotropy of the calculated maps. They are caused by errors in the localization and interpolation of the representative points.

To analyze the B-field maps, we present distributions of the radial displacements Δr of the representative points and associated averaged B-fields $\overline{B_\varphi}$ for each selected shot [see Fig. 12(a)]. We observe weak or no deflections within the $\sim$3 mm radius R_{cs} for all three shots, which means that there are almost no magnetic fields near the axis. In shot Nos. 2404 and 2420, the averaged B-fields are monotonically increasing with the radius r_C, and so, interestingly, the current sheath must be quite broad and spread under the D-grid from 3 to at least 13 mm radius. In shot No. 2408 [cf. Fig. 11(c)], the averaged B-field has a peak at the $\sim$7 mm radius, which is followed by a spread of points at a greater distance because the displacements become asymmetric. The maximal deviation error of the point localization in the deflectograms is $\sim$0.75 mm, which corresponds to the $\sim$15% uncertainty by Eq. (6). It is comparable to the error due to the asymmetry of the B-fields and due to the position of the pinhole, which is in shot Nos. 2404 and 2408 located slightly off-axis. However, the presented values of the averaged

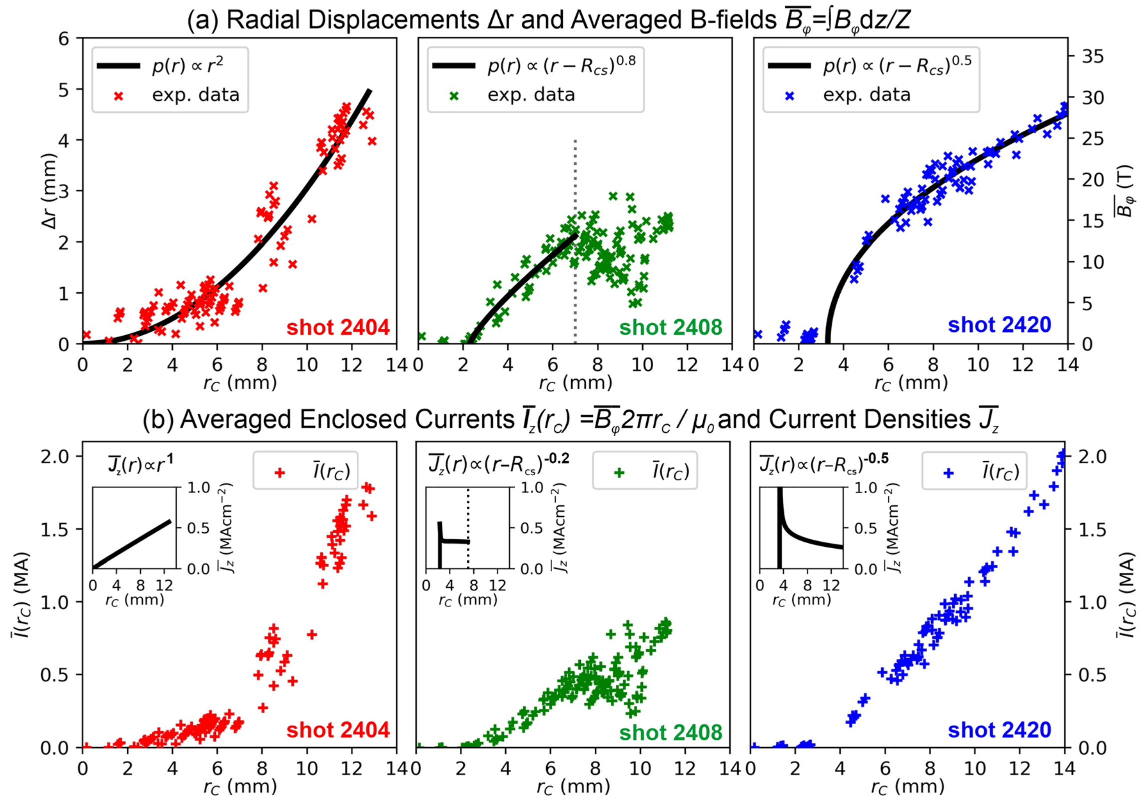

FIG. 12. Radial distributions of **(a)** radial displacements Δr together with corresponding averaged B-fields $\overline{B_\varphi}(r_C)$, and **(b)** averaged enclosed currents $\overline{I}(r_C) = \overline{B_\varphi}(r_C)2\pi r_C/\mu_0$ vs r_C. Averaged B-fields $\overline{B_\varphi}$ are calculated using Larmor orbit approximation and Eq. (5) from the radial displacements Δr of the retrieved points of the D-grid shadow in the deflectograms shown in Fig. 11. To approximate the radial profiles of the averaged current densities $\overline{J_z}(r_C)$, we fit the experimental data by a power function $p(r) \propto (r - R_{cs})^n$, similarly to the analysis in Sec. III D. The radius R_{cs} is 0, 2.3, and 3.3 mm for shot Nos. 2404, 2408, and 2420, respectively.

B-field can be even higher (by ~40%) because deuterons with higher energies may contribute to some parts of the deflectogram.

Moreover, we can fit the experimental data by a power function $p(r) \propto (r - R_{cs})^{n_B}$ and discuss the exponent n_B, similarly to the analysis in Sec. III D. We assume that R_{cs} is equal to 0, 2.3, and 3.3 mm for shot Nos. 2404, 2408, and 2420, respectively. In shot No. 2408, we consider only the undispersed data within the radius $r_C \leq 7$ mm. The estimated exponents n_B of the averaged B-fields in the shots are $n_B^{2404} = 2$, $n_B^{2408} = 0.8$, and $n_B^{2420} = 0.5$.

For better comprehension of a role of these exponents, we outline the averaged current density profiles $\overline{J_z}$ [insets in Fig. 12(b)] using the Ampere's law in cylindrical coordinates $\overline{J_z} = (\partial r\overline{B_\varphi}(r))/(\mu_0 r \partial r)$, where we characterize the averaged B-fields $\overline{B_\varphi}(r)$ by their fits. We see that the power-law exponents n_B for the selected shots represent an increasing ($n_B^{2404} - 1 > 0$), almost constant ($n_B^{2408} - 1 \approx 0$), and decreasing ($n_B^{2420} - 1 < 0$) averaged current density profiles with respect to the radius r_C. Interestingly, only in shot 2404, the averaged current density is gradually increasing. Unlike in the other two, in this shot, the D-grid was supported by 4 wires at its edge instead of a single wire in the center. These wires interfered with the imploding plasma and so influenced the current sheath. Nevertheless, the presented experimental data do not have sufficient spatial resolution for more detailed study of the averaged current densities. While such measurements are among our future goals, the outlined profiles should rather provide qualitative information about $\overline{J_z}(r_C)$. For example, because of the decreasing gradients of the B-field profiles in shot Nos. 2408 and 2420, the averaged current density $\overline{J_z}(r_C)$ must be decreasing and have a maximum near the current sheath radius R_{cs}. However, it is difficult to correctly estimate its position and height.

In contrast, we can quite easily derive the radial distributions of the averaged currents $\overline{I}(r_C)$ enclosed within the radius r_C [see main figures in Fig. 12(b)] from the measured experimental data by a relation $\overline{I}(r_C) = \overline{B_\varphi}(r_C)2\pi r_C/\mu_0$. Maximal values of $\overline{I}(r_C)$ in the featured shots are in agreement with the results in Sec. IV B. While the averaged current density $\overline{J_z}(r_C)$ is coupled with detailed current structure, the averaged enclosed current $\overline{I}(r_C)$ is inherently cumulative and provides less detailed information.

D. Mapping (r-z) profile of the azimuthal B-fields in shot 2420 using numerical simulations

Deflectograms reflect only the radial distribution of path-integrated B-fields $\int B_\varphi dz$, but provide no information about the axial profile of the B-fields. To estimate the actual B-field distribution, we need some knowledge of the complementary axial profile from a radial line of sight.

In the case of shot 2420, we have obtained this information by recording side-view SXR-emission images of the Z-pinch plasma via a multichannel plate (MCP) detector. Figure 13(a) shows an imploding Z-pinch plasma during the ion imaging. We suppose that an inner boundary of the current sheath has a similar parabolic shape as the depicted plasma instability, which we consider azimuthally symmetric. We set this the parabolic shape (but not the radial position) of the plasma boundary $r_{cs}(z)$ as an input parameter into the ion-tracking code.

We have tested various distributions of azimuthal B-fields $B_\varphi(r, z)$ and compared the simulated ion displacements from simulations with the measured D-grid displacements. In Fig. 13(b), we present a simulated D-grid pattern (red), which best fits not only the retrieved part of the D-grid shadow (blue) but the whole deflectogram of shot No. 2420 in Fig. 13(c). The artificial D-grid shadow is in good agreement with the experimental data. Discrepancies between the synthetic and observed D-grid pattern are given by the azimuthal asymmetry of B-fields in the experiment. For a better comparison of the whole picture, a complete synthetic deflectograms is illustrated in Fig. 13(d) revealing both the distorted D-grid shadow and an undistorted shadow of the cathode.

We propose numerical solution for $B_\varphi(r, z)$ given by a relation

$$B_\varphi(r < R_p, z) = \frac{\mu_0 I_p}{2\pi R_p}\left(\frac{r - r_{cs}(z)}{R_p - r_{cs}(z)}\right)^{m_B}, \tag{7}$$

where the I_p is total pinch current at the pinch radius R_p. From Sec. IV C, we know that the distribution of the averaged B-fields $\overline{B_\varphi}(r_C) = \int B_\varphi \mathrm{d}z/Z$ is proportional to $(r - R_{cs})^{n_B}$, where the exponent $n_B \approx 0.5$. While calculating $\overline{B_\varphi}$ from the path integral $\int B_\varphi \mathrm{d}z$, we assume a constant B-field height $Z = 10$ mm. However, if the B-fields are axially perturbed, the height of the actual B-fields depends on the radius. According to Fig. 13(a), it becomes smaller than the D-grid-cathode distance Z closer to the axis. To compensate this effect while keeping the $\int B dz$ profile given by the measured displacements, the cross-sectional profile of the actual B-fields $B_\varphi(r, z = \text{const.})$ must be steeper than the averaged B-field profile [cf. Fig. 14(a)]. Therefore, we estimate the exponent of m_B in Eq. (7) to be $\approx$0.2.

As seen from Fig. 14(a), the retrieved part of the D-grid pattern displays the distortions ranging from the radius of $R_{cs} = r_{cs}(z = 4.7\text{ mm}) \approx 3.3$ up to $\sim$14 mm. The pinch radius R_p cannot be smaller than 14 mm, because in that case there would be a peak in the averaged B-field profile, which we do not observe. However, the low estimate of the pinch radius R_p in Sec. IV B is 1 mm farther, that is, at 15 mm. It is caused by the $\sim$0.4 mm thickness of the D-grid edge because the retrieved points of the D-grid pattern refer only to its square openings, but the synthetic deflectogram also manifests the displacements around the circular D-grid edge shadow. Therefore, we determine a minimal pinch radius of $R_p = 15$ mm, which results in the low estimate of the 2.5-MA pinch current I_p and agrees with the displacement of the D-grid edge, measured in Subsection IV B.

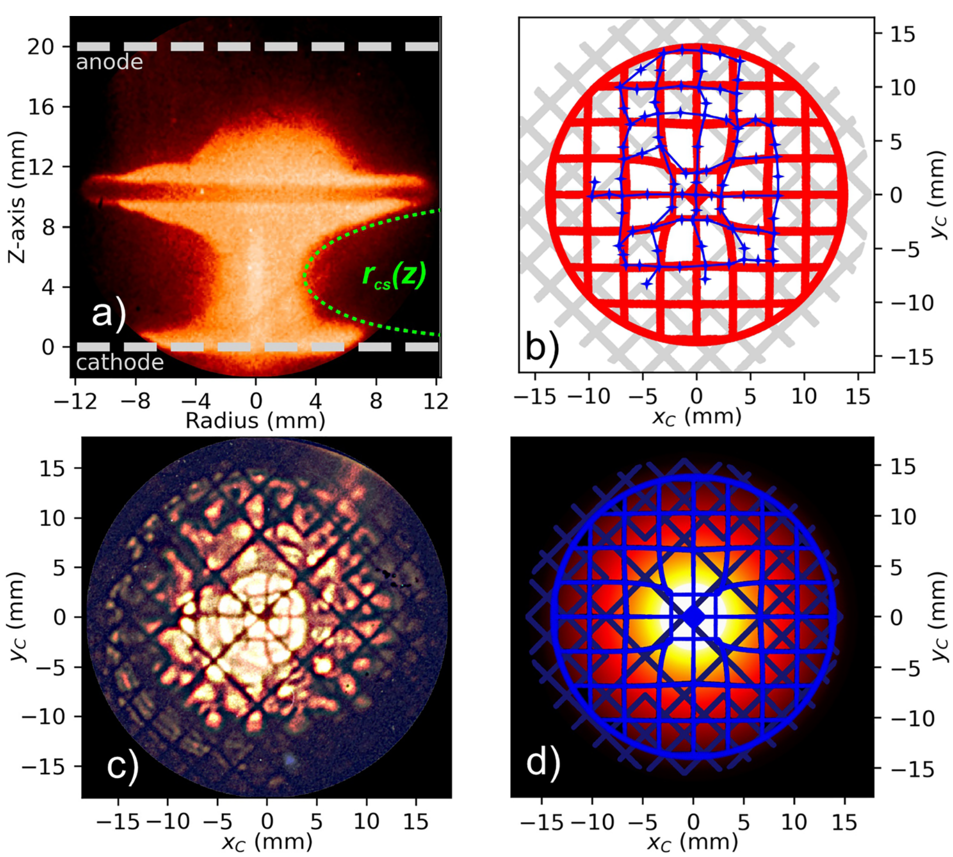

FIG. 13. **(a)** SXR image of the imploding plasma recorded by a multichannel plate (MCP) detector in the shot No. 2420 during ion imaging. We set the shape of the imploding plasma $r_{cs}(z)$ as an input parameter to our numerical code to characterize the current sheath perturbation. **(b)** Comparison of the simulated D-grid pattern with the retrieved part of the distorted D-grid from the experimental data; **(c)** experimental and **(d)** synthetic deflectograms of the distorted D-grid and the undistorted cathode mesh with the representative points.

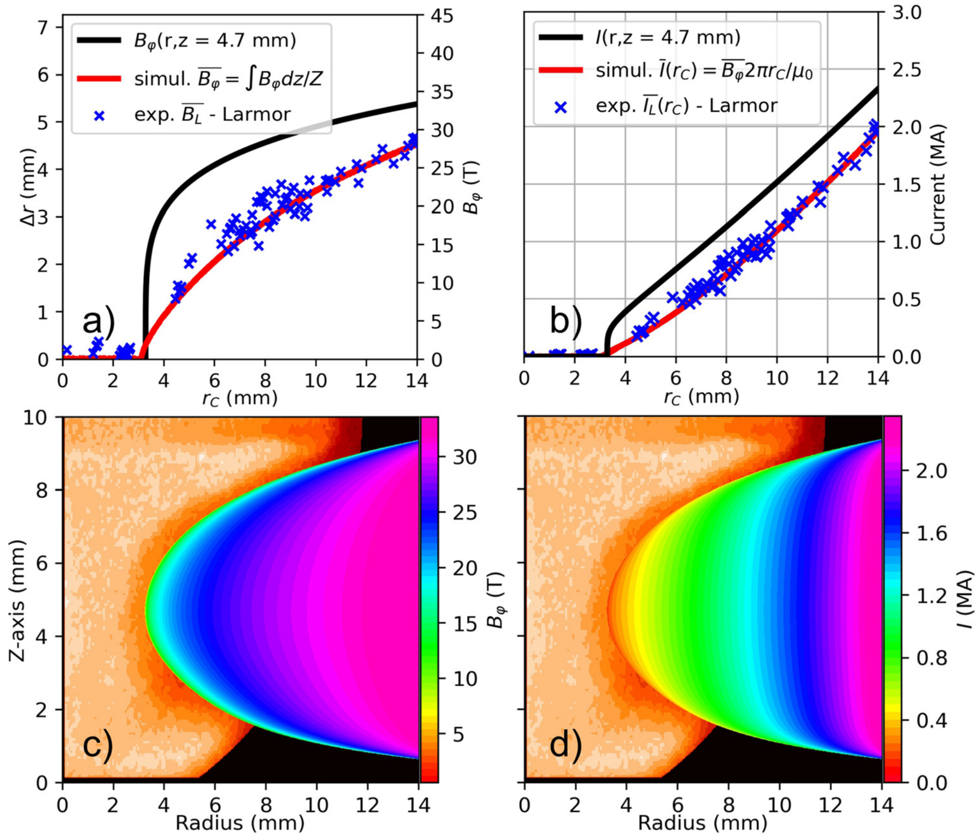

FIG. 14. Cross-sectional profiles of **(a)** the azimuthal B-fields $B_\varphi(r, z = 4.7\,\text{mm})$ and **(b)** the current $I(r, z = 4.7\,\text{mm})$ are compared to the averaged $\overline{B_\varphi}(r_C)$ and $\overline{I}(r_C)$. Topological (r-z) maps of **(c)** B-field $B_\varphi(r, z)$ and **(d)** current $I(r, z)$, which we obtain as a fitting input distributions from the numerical simulations reconstructing the experimental deflectogram in shot No. 2420.

Due to the abrupt increase in the current profile near the radius $R_{cs} \sim 3.3$ mm in Fig. 14(b), the current density must be high ($\geq 10^1$ MA cm^{-2}) at the thin layer of the current sheath front (roughly ~0.5 mm). With increasing radius, the B-fields slope become more steady, and thus, the estimated current density falls to 10^{-1} MA cm^{-2}. However, detailed information about the current density is not possible because it relies heavily on a specific fit of the B-fields. The uncertainty of this fit is caused by the uncertainty determination of the radial position of the onset of the ion deflections [see Fig. 14(a) between 3 and 4 mm], which cannot be determined precisely from the experimental deflectogram of the shot No. 2420.

From our simulations, we obtain 2D (r-z) distributions of the local B-field $\overline{B_\varphi}(r, z)$ and the corresponding current $\overline{I}(r, z)$, which we set as the input to our simulations [see Figs. 14(a) and 14(b)]. As seen, the current sheath is not only perturbed but also stretched along the whole interval from R_{cs} to R_p ($3.3 - 15$ mm). Therefore, the both B-field and current in the cross section of the current layer are monotonically increasing [see Figs. 14(c) and 14(d)]. If we would consider $R_p = R_p(z)$, then the strengths of the path-integrated B-fields would decrease from the minimal $R_p(z)$ and synthetic deflectogram would not reflect the ion displacements. Therefore, we set the outer boundary of the current sheath as a straight line and the pinch radius R_p = 15 mm applies for all z. We conclude the current sheath must be wide, stretched and situated in the region of the plasma with lower temperature (not glowing in SXR).

V. CONCLUSIONS AND FUTURE PLANS

We proved that ion deflectometry is a feasible diagnostic method for measuring the magnetic fields in Z-pinch plasmas. We have found a novel method for ion backlighting of the deflectometry grid. For our B-fields measurements, we have employed multi-MeV deuteron beams accelerated after disrupting the Z-pinch current in the unstable plasma neck. Thus, we performed the first Z-pinch-driven ion deflectometry experiments. We have developed the unique experimental setup with a fiducial deflectometry grid (D-grid) inserted in-between the electrodes. The D-grid did not suppress the ion emission near the anode while not severely affecting the plasma implosion below the D-grid. On the other hand, the D-grid stalk prevented deuterons to be accelerated between the D-grid and the cathode. Otherwise, they would not pass the D-grid, blur the deflectogram, and lower its contrast. Therefore, we could utilize these ions to map the azimuthal B-fields in the plasma below the D-grid, which manifested their deflections.

We obtained images of the D-grid shadow imprinted in the deflected ion beams, which displayed the radial profile of the path-integrated B-field $||\int \boldsymbol{B} \times \mathrm{d}\boldsymbol{L}||$ in an area within a 15 mm radius from the axis. The experimental setup with a distant pinhole camera set a minimal observation angle of the detected ions. It allowed us to estimate the path-integrated B-fields $\int B_\varphi dz$ using the Larmor orbit approximation.

We presented the experimental data from several shots with the different quality of the obtained information. Even from the low-quality deflectograms, we estimated the minimal value of the total pinch current within a 15 mm radius from the axis. It reached up to 2.5 MA for a few shots, which was close to the 2.7-MA total circuit current during the stagnation.

Using our numerical path-tracking simulations for various azimuthal B-fields, we have investigated how the deflectogram structure reflects specific B-field profiles. In three selected shots, where the distorted D-grid pattern was recognizable, we retrieved the (x-y) maps of the averaged B-fields. They revealed that, within the 3 mm distance from the axis, there were almost no B-fields at the time of ion imaging. Moreover, we were able to estimate the profile of the averaged current density for these selected shots.

In shot No. 2420, we captured the SXR image of the imploding Z-pinch plasma during the ion emission and utilized it to infer the axial shape of the current sheath. Using our numerical simulations, we found the fit for the experimental data and, as a result, reconstructed a topological (r-z) map of the local B-fields. It implies from our results that the current sheath is quite broad ($\sim$12 mm) and that the majority of the current was flowing in the peripheral plasma, which does not glow in the SXR spectrum.

All these experimental data have been obtained during our first experimental campaign while testing the concept of the ion deflectometry measurements. We determined that beam energy estimation and the deflectogram contrast are critical elements of our topics that need improvement. Due to the high uncertainty in energy of the detected deuterons (between 2.3 and 4.7 MeV), we report only low estimates of the measured B-fields. In the future, we will solve this problem by using improved detector stack with optimized filters. Moreover, we are planning experiments employing D-grids with larger diameters to map B-fields farther from the axis and with lower supporting stalk to measure larger deflections. We will test the D-grids with different sizes of openings for better contrast and to investigate further the role of the D-grid transparency and the D-grid stalk for the ion acceleration mechanism. We will test several inverse D-grids with very low transparency, where the spacings are in the place of the connecting wires of the classical D-grid. Therefore, we can analyze the high-fluence structures in the low-fluence background instead of vice versa. We also plan to shorten the time of ion emission compared to the plasma implosion below the D-grid by manipulating the initial parameters of the experiment and the structure of the D-grid. Nevertheless, it will always be essential that we must not drastically suppress the emission of ions for the backlighting in any of these experiments.

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DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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